Plots of coefficients in the optimal Hamiltonian

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Libraries

```
library(lattice)
```

Reading the data files.

```
# log_all <- read.csv("gap_scaling_all_data.csv")

log_c <- read.csv("gap_scaling_coefficients.csv")

#log_fit <- read.csv("loglog_fit_parameters_logs.csv")



# dev_all <- read.csv("gap_scaling_N_over_x_all_data.csv")

dev_c <- read.csv("gap_scaling_N_over_x_coefficients.csv")

# dev_fit <- read.csv("loglog_fit_parameters_N_over_x.csv")



# pow_all <- read.csv("gap_scaling_N_power_x_all_data.csv")

pow_c <- read.csv("gap_scaling_N_power_x_coefficients.csv")

# pow_fit <- read.csv("loglog_fit_parameters_N_power_x.csv")
```

Short overview of the data.

```
summary(log_c)
```

```
##        N              family           x              kmax
##  Min.   :  10   Length   :1752   Min.   :1.000   Min.   :   1.0
##  1st Qu.: 160   N.unique :   3   1st Qu.:3.000   1st Qu.:  40.0
##  Median : 370   N.blank  :   0   Median :3.000   Median : 160.0
##  Mean   : 423   Min.nchar:   8   Mean   :3.189   Mean   : 227.6
##  3rd Qu.: 670   Max.nchar:  14   3rd Qu.:4.000   3rd Qu.: 347.0
##  Max.   :1000                    Max.   :4.000   Max.   :1000.0
##        k              c_k             c_k_per_subset
##  Min.   :  1.0   Min.   :-7.000e-08   Min.   :0.000000
```

```
##  1st Qu.: 28.0   1st Qu.: 1.755e-02   1st Qu.:0.000000
##  Median :116.0   Median : 7.809e-02   Median :0.000000
##  Mean   :160.6   Mean   : 2.329e-01   Mean   :0.000304
##  3rd Qu.:269.0   3rd Qu.: 2.434e-01   3rd Qu.:0.000000
##  Max.   :558.0   Max.   : 1.000e+00   Max.   :0.100000
```

```
summary(dev_c)
```

```
##        N              family          x               kmax
##  Min.   :  10   Length   :885   Min.   : 1.000   Min.   :   1.0
##  1st Qu.: 190   N.unique :  1   1st Qu.: 1.000   1st Qu.:  64.0
##  Median : 460   N.blank  :  0   Median : 2.000   Median : 130.0
##  Mean   : 468   Min.nchar:  3   Mean   : 3.737   Mean   : 249.1
##  3rd Qu.: 730   Max.nchar:  3   3rd Qu.: 6.000   3rd Qu.: 370.0
##  Max.   :1000                   Max.   :10.000   Max.   :1000.0
##        k              c_k             c_k_per_subset
##  Min.   :  1.0   Min.   :-7.000e-08   Min.   :0.0000000
##  1st Qu.: 48.0   1st Qu.: 5.615e-02   1st Qu.:0.0000000
##  Median :105.0   Median : 1.974e-01   Median :0.0000000
##  Mean   :151.8   Mean   : 3.842e-01   Mean   :0.0006007
##  3rd Qu.:221.0   3rd Qu.: 8.321e-01   3rd Qu.:0.0000000
##  Max.   :541.0   Max.   : 1.000e+00   Max.   :0.1000000
```

```
summary(pow_c)
```

```
##        N               family           x               kmax
##  Min.   :  10.0   Length   :1127   Min.   :0.1000   Min.   :   1.0
##  1st Qu.: 190.0   N.unique :   1   1st Qu.:0.6000   1st Qu.:  21.5
##  Median : 430.0   N.blank  :   0   Median :0.9000   Median : 113.0
##  Mean   : 452.3   Min.nchar:   3   Mean   :0.7601   Mean   : 220.3
##  3rd Qu.: 700.0   Max.nchar:   3   3rd Qu.:1.0000   3rd Qu.: 340.0
##  Max.   :1000.0                    Max.   :1.0000   Max.   :1000.0
##        k              c_k             c_k_per_subset
##  Min.   :  1.0   Min.   :-7.000e-08   Min.   :0.0000000
##  1st Qu.: 15.5   1st Qu.: 2.088e-02   1st Qu.:0.0000000
##  Median : 87.0   Median : 1.070e-01   Median :0.0000000
##  Mean   :142.2   Mean   : 3.017e-01   Mean   :0.0002232
##  3rd Qu.:237.5   3rd Qu.: 3.905e-01   3rd Qu.:0.0000000
##  Max.   :541.0   Max.   : 1.000e+00   Max.   :0.1000000
```

Visualization of c_k for different N and k_{max}.

The system size can be chosen. In the first plot also x is a parameter.

```
N_value <- 220 # 10:30:1000
x_value <- 3 # 1, 2, 3 or 4

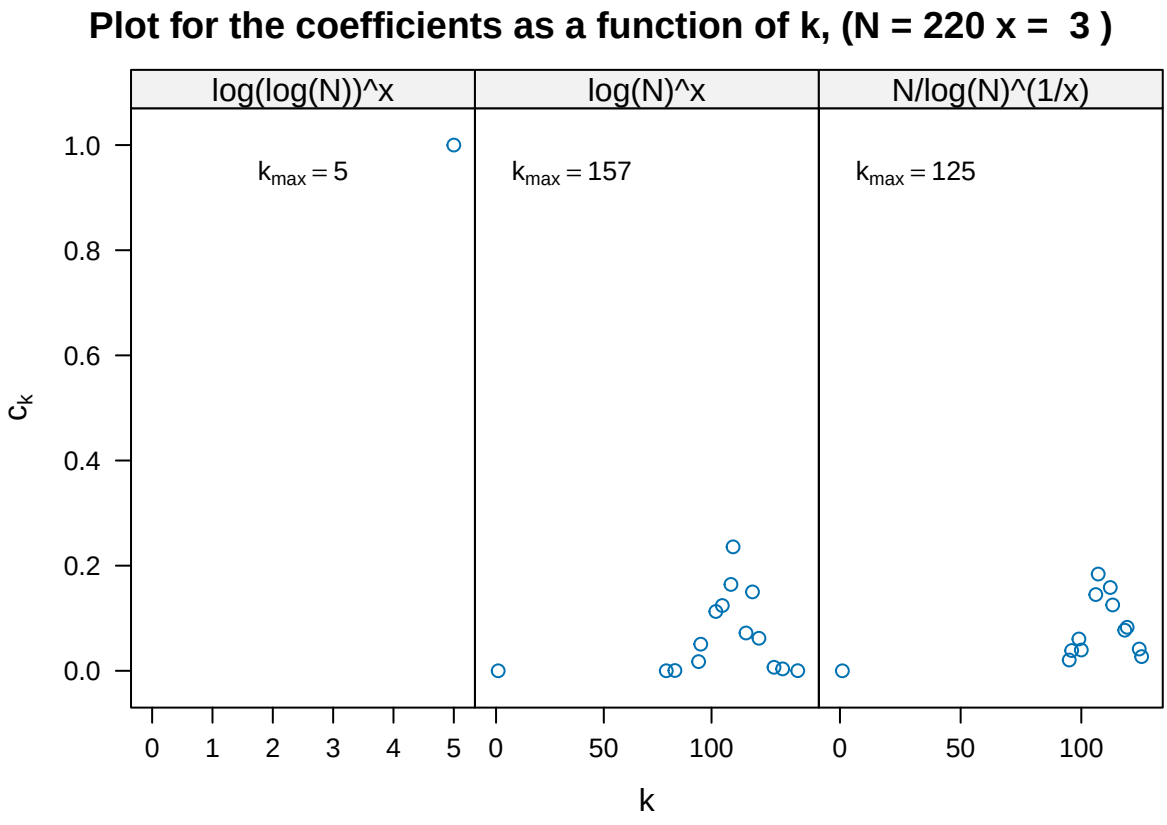
plot_data <- log_c[log_c$N == N_value & log_c$x == x_value, ]

xyplot(c_k ~ k | family,
       data = plot_data,
       xlab = "k",
       ylab = expression(c[k]),
       main = paste("Plot for the coefficients as a function of k, (N =", N_value, "x = ", x_value, ")"
       prepanel = function(x, y, ...) {
```

```
    list(
      xlim = c(0, max(x)),
      ylim = range(y)
    )
  },
  scales = list(
    x = list(relation = "free"),
    y = list(relation = "same"),
    alternating = FALSE,
    tck = c(1, 0)
  ),
  panel = function(x, y, subscripts, ...) {
    panel.xyplot(x, y, ...)

    k_max <- unique(plot_data$kmax[subscripts])

    panel.text(
      x = mean(range(x)) / 2,
      y = 0.9,
      labels =  bquote(k[max] == .(k_max)),
      pos = 3,
      cex = 0.8
    )
  })
```



```r
N_value <- 220 # 10:30:1000

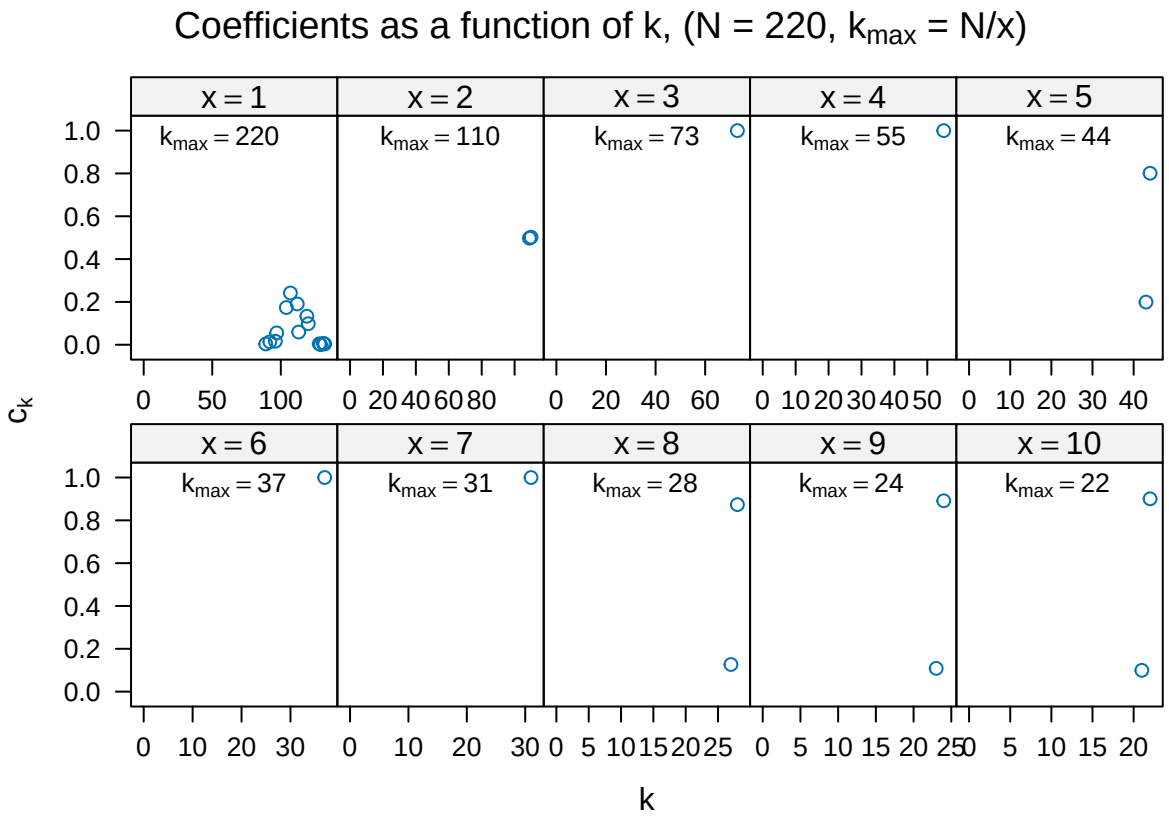
plot_data <- dev_c[dev_c$N == N_value, ]

plot_data$x <- factor(
  plot_data$x,
  levels = sort(unique(plot_data$x))
)

levs <- levels(plot_data$x)

strip_labels <- as.expression(
  lapply(levs, function(l) bquote(x == .(as.numeric(l))))
)

xyplot(c_k ~ k | x,
       data = plot_data,
       layout = c(5, 2),
       as.table = TRUE,
       xlab = "k",
       ylab = expression(c[k]),
       main = bquote("Coefficients as a function of k, (N = " * .(N_value) * ", " * k[max] * " = N/x)")
       strip = strip.custom(factor.levels = strip_labels),
       prepanel = function(x, y, ...) {
         list(xlim = c(0, max(x)), ylim = range(y))
       },
       scales = list(
         x = list(relation = "free"),
         y = list(relation = "same"),
         alternating = FALSE,
         tck = c(1, 0)
       ),
       panel = function(x, y, subscripts, ...) {
         panel.xyplot(x, y, ...)
         k_max <- unique(plot_data$kmax[subscripts])
         panel.text(
           x = mean(range(x)) / 2,
           y = 0.85,
           labels = bquote(k[max] == .(k_max)),
           pos = 3,
           cex = 0.8
         )
       })
```




```r
N_value <- 220 # 10:30:1000

plot_data <- pow_c[pow_c$N == N_value, ]

plot_data$x <- factor(
  plot_data$x,
  levels = sort(unique(plot_data$x))
)

xyplot(c_k ~ k | x,
       data = plot_data,
       xlab = "k",
       ylab = expression(c[k]),
       main = bquote("Coefficients as a function of k, (N = " * .(N_value) * ", " * k[max] * " = N^x)")
       strip = function(which.given, which.panel, var.name, factor.levels, ...) {
         if (which.given == 1) {
           panel_number <- which.panel[1]
           grid::grid.text(
             bquote(x == .(as.numeric(factor.levels[panel_number])))
           )
         } else {
           strip.default(which.given, which.panel, var.name, factor.levels, ...)
         }
       },
       as.table = TRUE,
       prepanel = function(x, y, ...) {
```

```
      list(
        xlim = c(0, max(x)),
        ylim = range(y)
      )
    },
    scales = list(
      x = list(relation = "free"),
      y = list(relation = "same"),
      alternating = FALSE,
      tck = c(1, 0)
    ),
    panel = function(x, y, subscripts, ...) {
      panel.xyplot(x, y, ...)

      k_max <- unique(plot_data$kmax[subscripts])

      panel.text(
        x = mean(range(x)) / 2,
        y = 0.7,
        labels =  bquote(k[max] == .(k_max)),
        pos = 3,
        cex = 0.8
      )
    })
```

